

Assignment 2

Last date: September 14, 2025

1. (0.5 marks each) How many positive integers less than 1000
 - (a) are divisible by 7?
 - (b) are divisible by 7 but not by 11?
 - (c) are divisible by both 7 and 11?
 - (d) are divisible by either 7 or 11?
 - (e) are divisible by exactly one of 7 and 11?
 - (f) are divisible by neither 7 nor 11?
 - (g) have distinct digits?
 - (h) have distinct digits and are even?
2. (1 mark) How many ways are there to distribute 12 distinguishable objects into six distinguishable boxes so that two objects are placed in each box?
3. Prove the following only via combinatorial argument (no marks will be provided otherwise) (2 marks each)
 - (a) $\binom{n}{k} < \binom{n}{k+1}$ where $k \leq n/2 - 1$ (assume n is even).
 - (b) $\sum_{k=0}^n \binom{n}{k} 2^k = 3^n$.
 - (c) $\sum_{r \geq 0} \binom{n}{r} \binom{m}{k-r} = \binom{n+m}{k}$.
 - (d) $\sum_{k=0}^n k^2 \binom{n}{k} = n(n+1)2^{n-2}$.
4. Prove via double counting (2 marks each)
 - (a) Let $\binom{[n]}{k} = \{S \subseteq [n] : |S| = k\}$ denote the set of all subsets of $[n]$ of size k . Suppose $\mathcal{F} \subseteq \binom{[n]}{k}$ be a family of k -subsets. Consider a new family $\mathcal{F}'$ defined as

$$\mathcal{F}' = \{S \subseteq [n] : |S| = k+1, S \supseteq T \text{ for some } T \text{ in } \mathcal{F}\}.$$

Prove that

$$|\mathcal{F}'| \geq \frac{n-k}{k+1} |\mathcal{F}|$$

- (b) At a party, the number of guests who shake hands an odd number of times is even.

5. Power Series, Generating Functions

- (a) (2 marks) How many ways can 1000 dollar be made using an unlimited number of coins of denominations 1, 2, 5, 10?
- (b) (2 marks) Find a recurrence relation (along with initial conditions) for the number of ways to completely cover a $2 \times n$ checkerboard with 1×2 dominoes.
- (c) (6 marks) Let c_n be the number of permutations on $[n]$ in which all cycles are of odd length.
- i. Show that c_n satisfies, for $n \geq 3$

$$c_n = c_{n-1} + (n-1)(n-2)c_{n-2}$$

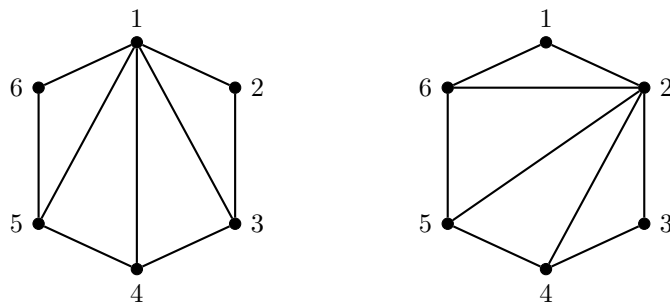
- ii. Give initial conditions and determine the closed form expression of the EGF of the above sequence.
- iii. Extract coefficient to show

$$c_n = \binom{n}{n/2} \frac{n!}{2^n} \text{ for even } n$$

and

$$c_n = \binom{n-1}{(n-1)/2} \frac{n!}{2^{n-1}} \text{ for odd } n$$

- (d) (6 marks) Let c_n be the number of strings of length n over $\{0, 1, 2\}$ that do not contain consecutive 2s. Write the recurrence relation for sequence c_n , derive the expression of its OGF and finally extract coefficient from it to get closed form expression for c_n .
- (e) (2 + 1 marks) Let c_n denote the number of ways to triangulate a convex polygon with n labelled vertices; that is, the number of ways to partition the polygon into triangles. The figure below shows two different triangulations of a hexagon.
- i. Write a recurrence relation for the sequence $c_1, c_2, \dots$, along with the initial conditions.
- ii. Determine c_n (you can use the facts proven in class without reproving).



6. Pigeonhole principle (2 marks each)

- (a) Five points are inside or on the boundary of an equilateral triangle of side length 2. Prove that there exist two points at distance at most 1 from each other.
- (b) Prove that any set of $n+1$ integers chosen from $\{1, 2, \dots, 2n\}$ contains two numbers one of which divides the other.
- (c) Any sequence of n^2+1 distinct numbers has a monotone subsequence (either decreasing or increasing) subsequence of length n .

7. Inclusion Exclusion Principle (2 marks)

- (a) Use the principle of inclusion and exclusion to determine the number of ways in which three women and their three spouses may be seated around a round table under the restriction that no woman sits beside her spouse (on either side).

8. Linear Algebra (2 marks each)

- (a) Show a family $\mathcal{F} \subseteq 2^{[n]}$ such that $|\mathcal{F}| = 2^{\lfloor n/2 \rfloor}$ and satisfies both the following conditions:
 - i. for all distinct $A, B \in \mathcal{F}$, $|A \cap B|$ is even,
 - ii. $|A|$ is even for all $A \in \mathcal{F}$.
- (b) Let $\mathcal{F} \subseteq 2^{[n]}$ be such that:
 - i. for all distinct $A, B \in \mathcal{F}$, $|A \cap B|$ is odd,
 - ii. $|A|$ is even for all $A \in \mathcal{F}$.

Show that $|\mathcal{F}| \leq n$.

(Hint: Assume $|\mathcal{F}| = n+1$. Show that $|\mathcal{F}|$ is odd by similar linear algebra technique as done in class. Then consider dual family $\mathcal{F}' = \{A^c | A \in \mathcal{F}\}$)